

CUSTODI

Think before you trust.

Verify before you act.

An on-device check that tells students whether a placement, internship or recruiter message is safe to act on — and shows exactly why.

TEAM · ANTRIKSH · DARSH · KUSHAGRI

A fake offer is designed to be believed

PS 5 — THE FAKE HR EMAIL · AS WORDED IN OUR BRIEF

"A student pastes a suspicious message and instantly receives:

1. Low, Medium or High risk.

2. Reasons why the message looks suspicious.

3. A clear next step.

4. A recommendation to verify suspicious real offers through the official placement cell."

Target users: college students seeking placements and college placement cells.

Specific warnings: OTP theft, ignored training and alert fatigue.

WHY "BE CAREFUL" ISN'T ENOUGH

— Training is forgotten under deadline pressure.

— Constant warnings get ignored — alert fatigue.

— Between receiving and verifying, the student has only the message.

From **hr.northwind.careers@gmail.com**[1]

Subject Congratulations — confirm your selection

Dear Candidate,

Congratulations! You have been shortlisted for the Graduate Analyst role at Northwind Analytics. To confirm your selection, please **share the OTP**[2] you receive on your registered mobile number **within 15 minutes**[3]. Failure to respond means your **offer will be cancelled**[3].

Regards, Talent Team

REAL CUSTODI RESULT: 65/100 · HIGH (SUSPICIOUS) · TEST LAB CASE M01

1 **Personal Gmail for official hiring**

+20 · impersonation

2 **Asks the student to share an OTP**

+35 · critical: always High

3 **Deadline plus a threat**

+10 · artificial urgency

CUSTODI: check first, then decide

Message received
email · SMS · WhatsApp · a link

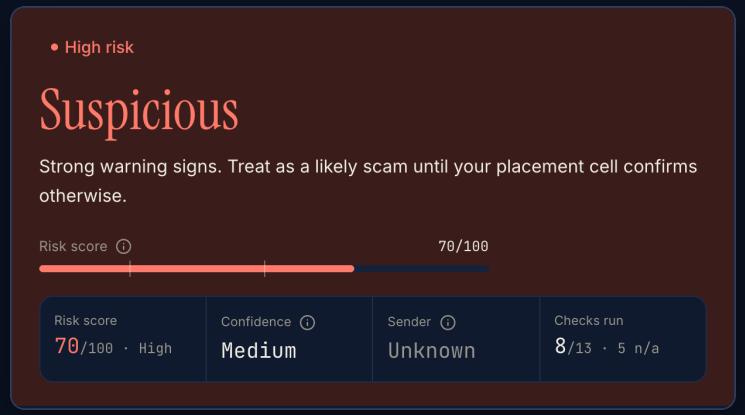
CUSTODI analysis
on the device, in seconds

Explainable assessment
risk, quoted evidence, every point

Recommended action
verify via the placement cell

01 MESSAGE SCANNER

Paste email, SMS or WhatsApp text. Get a risk score, the exact words behind it, and what to do next.



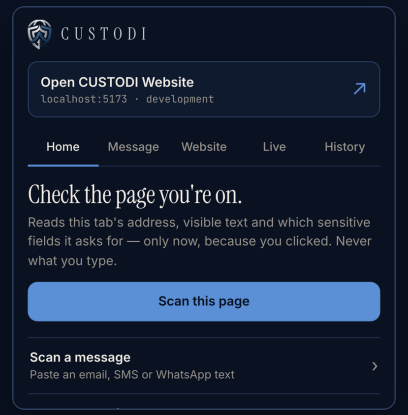
02 WEBSITE CHECKER

Reads an address without opening it. Flags structural tricks; unknown reputation is shown as “unavailable”.



03 CHROME EXTENSION

Scans the page you're on only when you click. Message and website checks in the popup.



CUSTODI helps students decide. It never guarantees a message is safe — or proves it is a scam.

One deterministic engine, shared by every surface

01 **Input**
Message, channel, optional sender + 3 answers (≤ 10,000 chars)

02 **Parse**
Split text, extract links and sender domain; normalise addresses

03 **Indicators**
13 checks · rules in English and common Hinglish

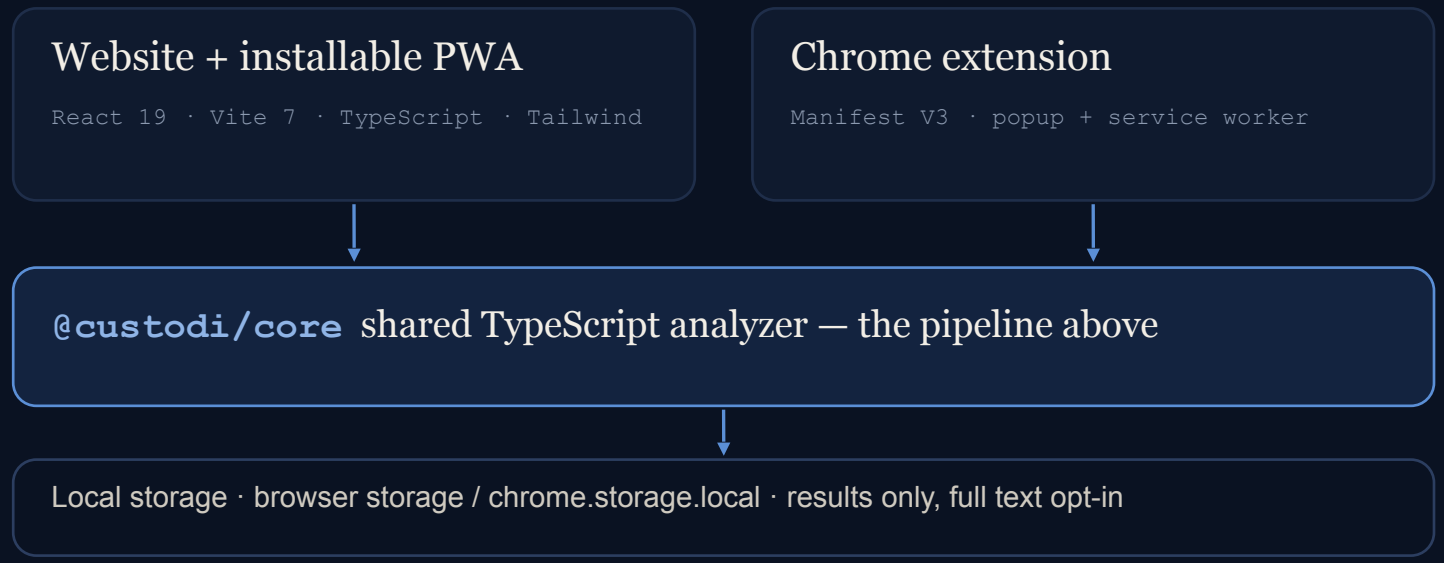
04 **Context & negation**
“never share”, “no fee”, quoted warnings, official portals → notes

05 **Scoring**
Weighted categories, each counted once, capped at 100

06 **Classification**
Low · Caution · Suspicious; critical override; confidence

07 **Report**
Quoted evidence, every point, next steps — saved locally

ARCHITECTURE · IMPLEMENTED TODAY



WHY THIS IS PRACTICAL

No mandatory AI API
Nothing to pay for per scan, nothing to leak.

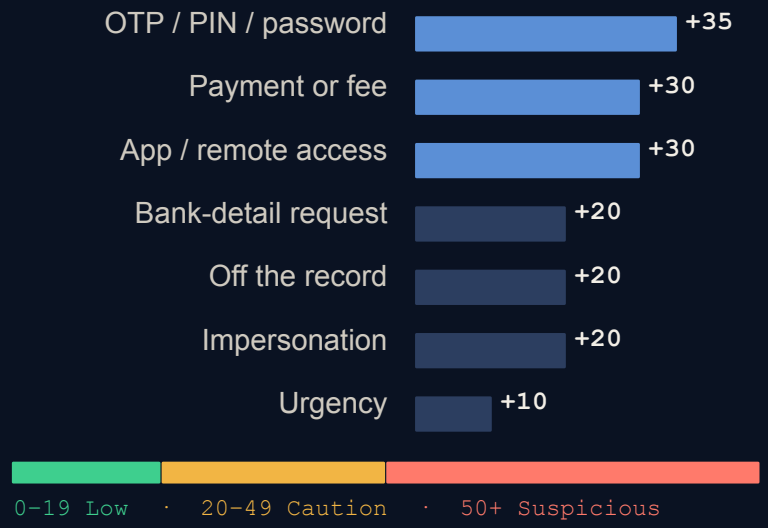
Predictable
Same input, same result — so it can be tested.

Low infrastructure
Static hosting; analysis runs in the browser.

One shared engine
Website, app and extension can't disagree.

Context changes the verdict — not just keywords

POINTS PER WARNING · EACH COUNTED ONCE



Critical override → High: OTP/PIN/password, remote access, .apk/.exe

WHATSAPP · BEFORE ANY OFFER

"...send your bank account number and IFSC code on WhatsApp to this number today..."

70 /100

High · Suspicious

+20 bank-detail request
+20 informal channel
+30 by message, no accepted offer

DEMO MESSAGE

EMAIL · AFTER ACCEPTING · GMAIL

"Now that you have accepted our offer, please email us your bank account number and IFSC..."

40 /100

Medium · Caution

+20 bank-detail request
+20 personal email for hiring
no +30: offer already accepted

TEST LAB R10

AFTER ACCEPTING · OFFICIAL PORTAL

"...add your bank account number and IFSC on the official onboarding portal... Do not email these details."

0 /100

Low

Request becomes a 0-point note: official portal, not asked back by message

TEST LAB M42

HOW CONTEXT & NEGATION WORK · REAL TEST CASES

Negation

"Never share your OTP" is advice, not a request → 0 (M14)

Not a fee

Stipends and salaries aren't fees; "no fee" stays calm (M15)

Quoted warning

Scam text quoted in a placement-cell alert → note (R07)

Official channel

Bank details for an official portal, not by message → note (M42)

LIMIT: CUSTODI READS THE WORDING — IT CANNOT CONFIRM THAT A PORTAL OR WEBSITE IS GENUINELY OFFICIAL.

Built, tested and runnable today

Technical

- React, Vite and TypeScript — mature, open tooling
- One shared analyzer for website and extension
- Chrome Manifest V3 extension
- Deterministic local analysis — no model to host

Operational

- No cloud infrastructure required
- No paid API, no per-scan cost
- Static hosting — deployment config included
- Plain-language results; no security training needed

Privacy

- Scans only when the student acts
- Never reads WhatsApp, SMS or email inboxes
- Analysed on the device; nothing uploaded
- Stored locally: results + short quotes; full text opt-in

VALIDATION EVIDENCE · VERIFIED IN THE CURRENT REPOSITORY

120/120
automated unit tests pass (Vitest)

83/83
Test Lab cases behave as designed · 49 TP · 34 TN · 0 FP · 0 FN

110
production page loads — 11 routes × 5 widths × 2 themes: 0 overflow, 0 console errors

Chrome
extension loaded and driven for real: scans, links, history — 0 errors

THE TEST LAB CASES ARE FICTIONAL AND WRITTEN BY US — THEY ARE NOT A REAL-WORLD ACCURACY CLAIM.

ILLUSTRATIVE MODEL · NOTHING HERE IS VALIDATED YET

Free for students. Funded by institutions.

Colleges & universities

pay an institutional subscription



CUSTODI

Students & job seekers

free core checks, private by default

Preventing even one student from losing money or identity documents can justify a college's subscription.

B2C · STUDENTS

Who Students and first-time job seekers

Gets Free message and website checks; extension

Pays Free core; optional premium features later

B2B · COLLEGES

Who Placement cells and universities

Gets Institution tools*, awareness campaigns, consented anonymised trends*

Pays Institutional subscription

B2B · RECRUITERS

Who Recruitment firms and placement partners

Gets Verification workflows*, applicant awareness tools*

Pays Enterprise licence · custom integrations

** Not built yet. No paying customers, partnerships, pricing or willingness to pay have been validated.*

Grows along four dimensions — from a working base

	WORKING TODAY	PLANNED NEXT	LONG-TERM POSSIBILITIES
Product	13-check message engine, website checker, extension, installable app · English + Hinglish	More rules and contexts, more message formats, better website analysis	More Indian languages; other browsers where feasible
Technical	Shared core, modular checks, fully local, static hosting	Optional, opt-in backend for institutional deployments	Scalable storage and analytics for consented, aggregate trends
Adoption	Public GitHub repo, installable app, unpacked extension	Placement-cell pilot · Chrome Web Store release	University partnerships and recruitment ecosystems
Geographic	Indian student recruitment scams	Wider job-seeker protection in India	Other languages and regions

Only the highlighted column exists today. Everything else is a plan, not a feature.

A fair comparison: what each approach is built for

	Spam filters	Antivirus / browser security	URL reputation checkers	Placement cell (manual)	CUSTODI
Built for recruitment scams	Not the focus	Not the focus	Not the focus	Yes	**Yes**
Reads what a message asks you to do	Partly	Not the focus	Not the focus	Yes	**Yes**
Checks web addresses	Often	Yes	Yes	Manually	**Structure only**
Known-bad reputation data	Yes	Yes	Yes	Partly	**No**
Explains each point	Rarely	Rarely	Partly	Yes	**Yes**
Context-aware bank / payroll reading	Not the focus	Not the focus	Not the focus	Yes	**Yes**
Instant, at any hour	Yes	Yes	Yes	No	**Yes**
Local-first — nothing uploaded	Varies	Varies	No	n/a	**Yes**

BASED ON EACH CATEGORY'S TYPICAL PURPOSE — INDIVIDUAL PRODUCTS VARY. CUSTODI COMPLEMENTS THESE TOOLS; IT DOESN'T REPLACE THEM.

Other tools ask “is this sender or link known to be bad?”
CUSTODI asks “what is this message asking me to do — and does that make sense here?”

From prototype to every placement cell

Now — Working prototype: website, installable app, extension and Test Lab — open on GitHub.

Next — Deploy the website, point the extension at it, submit to the Chrome Web Store.

Pilot — One placement cell: introduce CUSTODI at placement orientation and pin the link in official groups.

Measure — False alarms on genuine messages, scams reported to the cell, time taken to verify.

Scale — Recruiter lists verified by placement cells; more colleges and languages.

Intended impact: fewer students acting on fake offers before they verify — measured in the pilot, not assumed.

TEAM CUSTODI · ANTRIKSH · DARSH · KUSHAGRI



C U S T O D I

Think before you trust.

Verify before you act.

Because the cost of trusting the wrong message can be much greater than the time it takes to verify it.

github.com/govindanidarsh/Custodi

Open source · runs on your device · every example is fictional